Lili Ju
Curriculum Vitae

Department of Mathematics
University of South Carolina
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Education

Ph.D. in **Applied Mathematics**, 2002/05, Iowa State University

M.S. in **Computational Mathematics**, 1998/07, Chinese Academy of Sciences

B.S. in **Mathematics**, 1995/07, Wuhan University, China

Working Experiences

2013/01 – Present Professor
Department of Mathematics, University of South Carolina

2008/08 – 2012/12 Associate Professor
Department of Mathematics, University of South Carolina

2004/08 – 2008/08 Assistant Professor
Department of Mathematics, University of South Carolina

2002/09 – 2004/08 Industrial Postdoctoral Fellow
Institute for Mathematics and its Applications, University of Minnesota

Awards and Honors

- **SIAM Fellow**, Society for Industrial and Applied Mathematics, 2025.
- **World's Top 2% Scientists** by Stanford/Elsevier, 2023, 2024, 2025.
- **Plenary Speaker**, 45th SIAM Southeastern Atlantic Section Annual Meeting, 2023.
- **Plenary Speaker**, 12th China National Conference on Computational Mathematic, 2019.
- **ACM Gordon Bell Prize finalist** in High Performance Computing, 2016 (for largest and fastest 3D phase field microstructure coarsening simulations).

Research Interests

Numerical algorithms and analysis, scientific machine learning, computational geoscience, imaging and computer vision, high performance computing

Selected Research Grants

- ☐ **Department of Energy**, DE-SC0025527, *"DyGenAI: Dynamic Generative Artificial Intelligence for Prediction and Control of High-Dimensional Nonlinear Complex Systems"*, PI, 9/1/2024 – 8/31/2027, $450,000.
- ☐ **National Science Foundation**, DMS-2409634, *"Structure-Preserving Linear Schemes for the Diffuse-Interface Models of Incompressible Two-Phase Flows with Matched Densities"*, PI, 08/01/2024 – 07/31/2028, $294,473.
- ☐ **Department of Energy**, DE-SC0022254, *"Reliable and Efficient Machine Learning for Leadership Facility Scientific Data Analytics"*, PI, 09/01/2021 – 08/31/2024, $300,000.
- ☐ **National Science Foundation**, DMS-2109633, *"Maximum Bound Principle-Preserving Time Integration Methods for Some Semilinear Parabolic Equations"*, PI, 07/01/2021 – 06/30/2024, $150,000.

- ☐ **Department of Energy**, DE-SC0020270, *"Efficient and Scalable Time-Stepping Algorithms and Reduced-Order Modeling for Ocean System Simulations"*, PI, 09/01/2019 – 08/031/2022, $375,000.
- ☐ **National Science Foundation**, DMS-1818438, *"Study on Localized Exponential Time Differencing Methods for Evolution Partial Differential Equations"*, PI, 08/01/2018 – 07/31/2021, $150,000.
- ☐ **Department of Energy**, DE-SC0016540, *"Grid Generation, Coupling Strategies, and Spatially-Dependent Time Stepping for Ocean Tidal/Estuary Systems and Other ESM Components"*, PI, 09/01/2016 – 08/31/2019, $584,495.
- ☐ **National Science Foundation**, DMS-1521965, *"Fast and Stable Compact Exponential Time Difference Based Methods for Some Parabolic Equations"*, PI, 08/01/2015 – 07/31/2018, $200,954.
- ☐ **National Science Foundation**, DMS-1215659, *"Numerical Improvements, Mesh Adaptation and Parameter Identification for Parallel Finite Element Stokes Ice Sheet Modeling"*, PI, 08/01/2012 – 07/31/2015, $157,618.
- ☐ **Department of Energy**, DE-SC0008087-ER65393, *"Predicting Ice Sheet and Climate Evolution at Extreme Scales (PISCEES)"*, PI, 06/15/2012 – 06/14/2017, $348,420.
- ☐ **National Science Foundation**, DMS-0913491, *"Study on Algorithms and Applications of Centroidal Voronoi Tessellations"*, PI, 09/01/2009 – 08/31/2012, $180,000.
- ☐ **Department of Energy**, DE-FG02-07ER64431, *"New Grid and Discretization Technologies for Ocean and Ice Simulations"*, PI, 07/15/2007– 07/14/2011, $236,871.
- ☐ **National Science Foundation**, DMS-0609575, *"Some Problems on Analyses and Applications of Centroidal Voronoi Tessellations"*, PI, 08/01/2006 – 07/31/2009, $122,781.

Selected Publications

A total of over 180 articles in journals and conference proceedings and 1 book chapter
Google scholar record: 8104 citations and H-index 48 (as of December 4, 2025)

1. L. Ju, Q. Du and M. Gunzburger, Probabilistic methods for centroidal Voronoi tessellations and their parallel implementations, ***Parallel Comput.***, Vol. **28**, pp. 1477-1500, 2002.
2. Q. Du, M. Gunzburger and L. Ju, Constrained centroidal Voronoi tessellations for surfaces, ***SIAM J. Sci. Comput.***, Vol. **24**, pp. 1488-1506, 2003.
3. Q. Du and L. Ju, Approximations of a Ginzburg-Landau model for superconducting hollow spheres based on spherical centroidal Voronoi tessellations, ***Mathematics of Computation***, Vol. **74**, pp. 1257-1280, 2005.
4. Q. Du, M. Emelianenko and L. Ju, Convergence of the Lloyd algorithm for computing centroidal Voronoi tessellations, ***SIAM J. Numer. Anal.***, Vol. **44**, pp. 102-119, 2006.
5. J. Wang, L. Ju and X. Wang, An edge-weighted centroidal Voronoi tessellation model for image segmentation, ***IEEE Trans. Image Proc.***, Vol. **18**, pp. 1844-1858, 2009.
6. W. Zhao, G. Zhang and L. Ju, A stable multi-step scheme for solving backward stochastic differential equations, ***SIAM J. Numer. Anal.***, Vol. **48**, pp. 1369-1394, 2010.
7. W. Leng, L. Ju, M. Gunzburger, S. Price and T. Ringler, A parallel high-order accurate finite element nonlinear Stokes ice sheet model and benchmark experiments, ***J. Geophys. Res.***, Vol. **117**, F01001 (24 pages), 2012.
8. Q. Du, L. Ju, L. Tian and K. Zhou, A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models, ***Math. Comp.***, Vol. **82**, pp. 1889-1922, 2013.
9. J. Zhang, C. Zhou, Y. Wang, L. Ju, Q. Du, X. Chi, D. Xu, D. Chen, Y. Liu and Z. Liu, Extreme-scale phase field simulations of coarsening dynamics on the Sunway TaihuLight supercomputer, ***Proc. Int. Conf. High Perf. Comput., Network, Storage and Anal.*** (SC'16), Article #4 (12 pages), Salt Lake City, UT, USA, 2016. (Gordon-Bell Prize finalist)

10. X. Yang and L. Ju, Efficient linear schemes with unconditionally energy stability for the phase field elastic bending energy model, ***Comput. Methods Appl. Mech. Engrg.***, Vol. **315**, pp. 691-712, 2017.
11. X. Yang and L. Ju, Linear and unconditionally energy stable schemes for the binary fluid-surfactant phase field model, ***Comput. Methods Appl. Mech. Engrg.***, Vol. **318**, pp. 1005-1029, 2017.
12. H. Tian, L. Ju and Q. Du, A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization, ***Comput. Methods Appl. Mech. Engrg.***, Vol. **320**, pp. 46-67, 2017.
13. L. Ju, X. Li, Z. Qiao and H. Zhang, Energy stability and error estimates of exponential time differencing schemes for the epitaxial growth model without slope selection, ***Math. Comp.***, Vol. **87**, pp. 1859-1885, 2018.
14. T.-T.-P. Hoang, W. Leng, L. Ju, Z. Wang and K. Pieper, Conservative explicit local time-stepping schemes for the shallow water equations, ***J. Comput. Phys.***, Vol. **382**, pp. 152-176, 2019.
15. Q. Du, L. Ju, X. Li and Z. Qiao, Maximum principle preserving exponential time differencing schemes for the nonlocal Allen-Cahn equation, ***SIAM J. Numer. Anal.***, Vol. **57**, pp. 875-898, 2019.
16. Z. Wu, X. Wu, X. Zhang, S. Wang and L. Ju, Semantic stereo matching with pyramid cost volumes, ***Proc. Inter. Conf. Comput. Vis.*** (ICCV'2019), pp. 7484-7493, Seoul, Korea, 2019.
17. W. Leng and L. Ju, An additive overlapping domain decomposition method for the Helmholtz equation, ***SIAM J. Sci. Comput.***, Vol. **41**, pp. A1252-A1277, 2019.
18. T.-T.-P. Hoang, L. Ju, W. Leng and Z. Wang, High order explicit local time-stepping methods for hyperbolic conservation laws, ***Math. Comp.***, Vol. **89**, pp. 1807-1842, 2020.
19. Q. Du, L. Ju, X. Li and Z. Qiao, Maximum bound principles for a class of semilinear parabolic equations and exponential time-differencing schemes, ***SIAM Rev.***, Vol. **63**, pp. 317-359, 2021.
20. Q. Chen, L. Ju and R. Temam, Conservative numerical schemes with optimal dispersive wave relations: Part I. Derivations and analyses, ***Numer. Math.***, Vol. **149**, pp. 43-85, 2021.
21. X. Wu, Z. Wu, H. Guo, L. Ju and S. Wang, DANNet: A one-stage domain adaptation network for unsupervised nighttime semantic segmentation, ***Proc. IEEE Int. Conf. Comput. Vis. Pattern Recog.*** (CVPR'2021), pp. 15769-15778, 2021.
22. R. Lan, L. Ju, Z. Wang, M. Gunzburger and P. Jones, High-order multirate explicit time stepping schemes for the baroclinic-barotropic split dynamics in primitive equations, ***J. Comput. Phys.***, Vol. **457**, #111050 (25 pages), 2022.
23. L. Ju, X. Li and Z. Qiao, Generalized SAV-exponential integrator schemes for Allen-Cahn type gradient flows, ***SIAM J. Numer. Anal.***, Vol. **60**, pp. 1905-1931, 2022.
24. A. Gruber , M. Gunzburger, L. Ju and Z. Wang, Energetically consistent model reduction for metriplectic systems, ***Comput. Methods Appl. Mech. Engrg.***, Vol. **404**, #115709 (21 pages), 2023.
25. Y. Teng, Z. Wang, L. Ju, A. Gruber and G. Zhang, Level set learning with pseudo-reversible neural networks for nonlinear dimension reduction in function approximation, ***SIAM J. Sci. Comput.***, Vol. **45**, pp. A1148-A1171, 2023.
26. Y. Geng, Y. Teng, Z. Wang and L. Ju, A deep learning method for the dynamics of classic and conservative Allen-Chan equations based on fully-discrete operators, ***J. Comput.l Phys.***, Vol. **496**, #112589 (20 pages), 2024.
27. Z. Zhang, F. Bao, L. Ju and G. Zhang, Transferable neural networks for partial differential equations, ***J. Sci. Comput.***, Vol. **99**, #2 (25 pages), 2024.

28. D. Hou, L. Ju and Z. Qiao, Energy-dissipative spectral renormalization exponential integrator method for gradient flow problems, ***SIAM J. Sci. Comput.***, Vol. **521**, pp. A3477-A3502, 2024.
29. T. Lu, L. Ju and L. Zhu, A multiple transferable neural network method with domain decomposition for elliptic interface problems, ***J. Comput. Phys.***, Vol. **530**, #113902 (26 pages), 2025.
30. R. Lan, M. Liu and L. Ju, Robust error analysis of stabilized linear EMAC-SAV finite element schemes for the incompressible Navier-Stokes equations, ***Math. Comp.***, 2025. DOI: 10.1090/mcom/4087

Invited Talks

A total of over 290 invited talks given at conferences and academic institutions

- **Colloquium and Seminars**
  - Domestic: Georgia Tech., Univ. of Georgia, Univ. of Florida, Florida State Univ., Univ. of Minnesota, Carnegie Mellon Univ., Univ. of Pittsburgh, Penn State Univ., Columbia Univ., Clemson Univ., Auburn Univ., Univ. of Alabama, Iowa State Univ., Univ. of Maryland College Park, Univ. of Wisconsin Madison, Univ. of North Carolina at Chapel Hill, Purdue Univ., Ohio State Univ., Michigan State Univ., Michigan Tech., Univ. of Tennessee Knoxville, George Washington Univ., Georgia Mason Univ., Texas Tech., Sandia National Lab., Oak Ridge National Lab., Sandia National Lab., etc.
  - International: Ajou Univ., National Univ.of Singapore, Nanyang Technological Univ., Univ. of Hong Kong, Chinese Univ. of Hong Kong; Hong Kong Polytechnic Univ., Chinese Academy of Sciences, Peking University, Tsinghua Univ., Beijing Normal Univ., Renmin Univ. of China, Beihang Univ., Shanghai Jiao Tong Univ., Tongji Univ., Eastern China Normal Univ., Zhejiang Univ., Wuhan Univ., Huazhong Univ. of Sci. & Tech., Univ. of Sci.&Tech. of China, Shandong Univ., Ocean Univ. of China, Jilin Univ., Sichuan Univ., Univ. of Electronic Sci. & Tech. of China, Sun Yat-sen Univ., South China Univ. of Tech., Southern Univ. of Sci. & Tech., Lanzhou Univ., etc.
- **Conference Presentations:** SIAM Annual Meetings, SIAM-SEAS Annual Meetings, SIAM-CSE Meetings, SIAM-MDS Meetings, ICIAM Meetings, AMS-SES Fall and Spring Meetings, AIMS Conferences Series, etc.

Major Services

- **Professional Societies**:
  - Member of SIAM Committee on Committees and Appointments, 2024/01-Present.
  - President (2008-2009) & Secretary (2007-2008) of SIAM Southeastern Atlantic Section.
- **Editorial Boards**:
  - Editor-in-Chief (2026/01 – Present), Associate Editor (2021/10 – 2025/12), *Numerical Methods for Partial Differential Equations*
  - Associate Editor (2024/02 – Present), *Mathematics of Computation*
  - Editorial Board Member (2023/01 – Present), *Journal of Scientific Computing*
  - Associate Editor (2022/07 – Present), *Advances in Computational Science and Engineering*
  - Associate Editor (2021/07 – Present) , *Advances in Applied Mathematics and Mechanics*
  - Associate Editor (2021/01 – Present), *Numerical Math.: Theory, Methods and Applications*
  - Associate Editor (2021/06 – 2023/09) , *Advances in Continuous and Discrete Models*
  - Associate Editor (2012/01 – 2017/12), *SIAM Journal on Numerical Analysis*
- **University Services**:
  - Faculty Advisor of SIAM Student Chapter, University of South Carolina, 2006-2009.

- Member of 19 Ph.D. Dissertation Committees at USC.
- College level: Member of IMI Executive Committee 2012-2015.
- University level: USC General Education Work Team "Life-Long Learning", 2007; Member of University Committee of Tenure and Promotion, 2014.

- **Panels and Reviews**:
  - Panelists for 8 panels in Computational Mathematics Program and Career Program (2008, 2010, 2012, 2013, 2014, 2016, 2023, 2025), NSF Division of Mathematical Sciences.
  - Texas Tech University Mathematics Graduate Program Review, April 2025
  - Reviewers of many proposals for domestic and international agencies: NSF, DOE, Singapore National Research Foundation, Ministry of Education Research Grant of Singapore, Research Grants Council of Hong Kong, National Research Foundation of United Arab Emirates, etc.
  - External research evaluators for 21 Promotion and/or Tenure cases at peer institutions.
  - Referees for over 60 academic journals and conference proceedings since 2004.

- **Conference Organizations**:
  - Member of Organizing Committee, The Thirty-Second SIAM Southeastern Atlantic Section Annual Meeting, Orlando, FL, 2008
  - Co-chair of Organizing Committee, The Thirty-Third SIAM Southeastern Atlantic Section Annual Meeting, Columbia, SC, 2009
  - Chair of Organizing Committee, International Workshop on Computational and Applied Mathematics, Beijing, China, 2012.
  - Co-chair of Organizing Committee, The Ninth Graduate Student Miniconference in Computational Mathematics, Columbia, SC, 2018
  - Co-chair of Organizing Committee, The First International Conference on Mathematical Modeling and Numerical Methods, Qingdao, China, 2019.
  - Member of Organizing Committee, The Second International Conference on Mathematical Modeling and Numerical Methods, Beijing, China, 2024.
  - Organizers/Co-organizers of mini-symposia: SIAM-SEAS 2006, SIAM-CSE 2007, SIAM-SEAS 2008, SIAM-AN 2008, ICIAM 2011, SIAM-SEAS 2012, SIAM-SEAS2014, ICIAM 2015, SIAA-SEAS 2017, ECM2017, SIAM-SEAS 2018, SIAM-SEAS 2021, SIAM-MDS 2022, SIAM-SEAS 2023, etc.

Research Supervision

- **Ph.D. Students**: Shuting Wang (Ph.D., USC/Math, current), Sabrina Rashid (Ph.D., USC/Math, current, co-supervisor), Anu Vasudevan (Ph.D., USC/Math, current), Yuwei Geng (Ph.D., USC/Math, current, co-supervisor), Yuankai Teng (Ph.D., USC/Math, 2023/8), Xinyi Wu (Ph.D., USC/CSE, 2022/8, co-supervisor), Zhenyao Wu (Ph.D., USC/CSE, 2022/8, co-supervisor), Shuai Yuan (Ph.D., USC/Math, 2020/8, co-supervisor), Chenfei Zhang (Ph.D., USC/Math, 2019/8), Youjie Zhou (Ph.D., USC/CSE, 2015/8, co-supervisor), Yu Cao (Ph.D., USC/CSE, 2013/8, co-supervisor), Xiao Xiao (Ph.D., USC/Math, 2012/5), Li Tian (Ph.D., USC/Math, 2009/5)

- **M.S. Students**: Meshack Kiplagat (M.S., USC/Math, 2013/12), Rosalia Tatano (M.S., USC/Math, 2013/5), Jing Liu (M.S., USC/Math, 2010/8), Xiao Xiao (M.S., USC/Math, 2010/5), Wensong Wu (M.S., USC/Math, 2007/8),

- **Postdoctoral Associates**: Rihui Lan (2020/5-2022/11), Anthony Gruber (2021/1-2022/8), Xucheng Meng (2018/9-2019/8), Xiao Li (2018/8-2019/8), Huadong Gao (2018/3-2018/12), Thi-Thao-Phuong Hoang (2017/1-2018/8), Jianfang Lu (2016/7-2018/8), Hao Tian (2013/7-2016/1), Tong Zhang (2013/8-2015/8), Xiaoran Shi (2012/7-2013/7), Huai Zhang (2008/9-2010/3)

- **Visiting Scholars/Students**: 15